



CORAL GARDENERS

UC San Diego

Team Coral Gardeners

University of California San Diego

Place-Specific Reef Fish Identification for Coral Restoration

Milestone Report

Spring 2026

Project Sponsor

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Introduction

This project focuses on developing a place-specific computer vision pipeline for reef fish identification to support coral restoration and reef biodiversity monitoring. The original project proposal outlined a two-stage pipeline involving fish detection followed by classification using geographically relevant datasets. As the quarter progressed, the project evolved into a broader evaluation and refinement framework involving multiple segmentation and detection approaches, including RF_DETR, SAM3, and BioCLIP classification. At this stage of the project, the team has successfully built an end-to-end pipeline, evaluated multiple segmentation and classification models, identified major failure cases, and improved system performance through model experimentation. This report summarizes the project's progress toward the MVP, completed milestones and deliverables, remaining milestones for the rest of the quarter, individual team member contributions, and how instructor and sponsor feedback influenced the direction and prioritization of the project.

Addressing Feedback

The primary feedback from our original proposal centered on a lack of clarity around team technical expertise and how responsibilities were distributed across members. In the initial proposal, responsibilities were broadly scoped and loosely assigned, making it unclear who owned which technical decisions or deliverables. To address this, we restructured the project around each member's specific technical strengths. This restructuring directly accelerated our progress. By matching team members to the components best suited to their expertise, we were able to evaluate multiple segmentation and classification models in parallel rather than sequentially, ship a working end-to-end pipeline, and complete all three MVP goals despite the unexpected discovery of domain shift as a major failure case.

Aleksa coordinated milestone progression and led end-to-end pipeline architecture, and final system reproducibility. Kaleigh led segmentation experimentation and evaluation across RF_DETR, SAM3, along with the SAM3-to-BioCLIP integration that established our baseline end-to-end pipeline. Ewan led dataset engineering, reproducibility workflows, and preprocessing experimentation targeting the underwater image quality bottleneck. Marlyn led YOLO model refinement, and comparative evaluation across multiple models and datasets. Vibusha led the sourcing the multiple reef imagery datasets used across our experiments, dataset preparation, buoy annotation, and augmentation workflows that simulate underwater environmental conditions. This clearer ownership model also

produced more pointed feedback cycles, since each update could be directed at the specific person responsible for that component.

Group Management

Due to the unexpected discovery of the major failure case of domain shift, we did not strictly abide by our initial project timeline and focused our efforts on evaluating multiple models rather than committing to one model and pipeline as described in the original plan. Furthermore, our project sponsor had limited evaluation data for us to comprehensively test our approach and simply commit to one mode. As a result, the accomplished list of milestones for each team member is different than the initial project plan as most of our progress was made in experimenting with multiple models and approaches.

Aleksa Popovic: Assigned to machine learning model integration, pipeline architecture, and project coordination. Aleksa focuses on integrating segmentation and classification workflows into a functioning end-to-end pipeline and organizing milestone progression. He also helps coordinate experiments involving RF_DETR and SAM3 segmentation and manages integration between different pipeline components.

Expertise: Aleksa brings strong Python and ML pipeline experience spanning data preprocessing, model training, and evaluation, along with experience of structuring clean, modular projects in team settings. Due to this experience, he coordinated milestone progression, aligned experimental tracks across team members, and ensured pipeline components integrated cleanly into a single deliverable. His computer vision background from the ScreenSense project covers image preprocessing, dataset building, FFT-based feature extraction, and EXIF metadata analysis, which directly supports his work on the end-to-end pipeline architecture and comparative evaluation of RF_DETR and SAM3 under domain shift.

Accomplishments: Aleksa contributed to the machine learning model integration and pipeline architecture throughout the project. He set up a more fine-tuned pipeline of detection and classification stages and evaluated various segmentation models. He generated various evaluation metrics to display tradeoffs between various models.

- Built and refined the end-to-end segmentation and classification pipeline architecture
- Evaluated multiple segmentation approaches, including RF_DETR and SAM3, under domain shift conditions
- Generated evaluation metrics and comparison graphs to analyze tradeoffs between segmentation models

Kaleigh Edusada: Assigned to segmentation experimentation and evaluation, specifically with YOLO and SAM3 computer vision models. She accomplished generating evaluation metrics and comparing segmentation performance across different datasets. She also did an integration of segmentation and classification with SAM3 and BioCLIP to get a baseline, functional end-to-end pipeline. Overall, Kaleigh contributes to baseline comparisons and performance analysis under domain shift conditions, specifically in our segmentation phase of the pipeline, alongside the integration for segmentation and classification.

Expertise: Kaleigh brings strong Python and machine learning fundamentals, with hands-on experience in data analysis, model evaluation, and the integration work required to connect ML components into a functional end-to-end pipeline. Her skill set spans dataset handling, metric generation, performance benchmarking, and comparative analysis across machine learning models. This systems-level thinking directly supports her ownership of segmentation experimentation across multiple models and ensuring SAM3-to-BioCLIP integrate cleanly.

Accomplishments: Kaleigh contributed heavily to evaluating segmentation models, including RF_DETR and SAM3. She created evaluation metrics and heavily contributed to establishing a functional end-to-end reef fish identification pipeline. She worked on the baseline integration of SAM3 with BioCLIP, where it will save the cropped images from segmentation and using those cropped images, BioCLIP will classify the species based on the location, where it then was fine-tuned by Aleksa.

- Conducted extensive segmentation experimentation using RF_DETR, SAM3, and baseline YOLO approaches
- Integrated SAM3 segmentation with BioCLIP classification to establish the initial end-to-end inference workflow
- Evaluated segmentation robustness and performance degradation under domain shift conditions

Ewan Shen: Primarily contributes to dataset engineering and reproducibility workflows. He organizes and cleans reef datasets, prepares training-ready datasets, annotates training data, and documents workflows for reproducibility. He is also developing preprocessing and evaluation pipelines to experiment with evaluation image processing. He will test and evaluate various preprocessing methods to choose a best performing technique to integrate into the final pipeline.

Expertise: Ewan has research experience running computer vision applications on RISC-V systems and project experience applying YOLO to estimate microplastic ingestion from household object imagery, particular focus on preprocessing techniques that facilitate

object detection. This expertise directly supports his work organizing, cleaning, and validating reef fish datasets, building reproducible preparation workflows, and leading the preprocessing experimentation that targets the underwater image quality bottleneck identified in our BioCLIP evaluation.

Accomplishments: Ewan organized and cleaned reef datasets, prepared training-ready data, contributed to our annotated dataset, and documented workflows to support reproducibility.

- Organized, cleaned, and validated reef fish datasets for training compatibility
- Contributed to annotated training datasets and reproducible dataset preparation workflows
- Contributed to initial community fish detector model evaluation and metric analysis

Marlyn Arque Rupa: Primarily focuses on model refinement and evaluation of existing machine learning workflows within the reef fish identification pipeline. Her responsibilities include improving segmentation and classification performance through experimentation, model refinement, robustness analysis, and comparative evaluation across multiple models and datasets. Marlyn worked extensively on analyzing prediction quality.

Expertise: Marlyn has strong Python skills and hands-on experience with Segment Anything models, having previously built an end-to-end web application that integrates SAM-based and other imaging models. She also has experience with image analysis, team-based development, and rapidly onboarding new technologies. This background directly supported her work training and evaluating the baseline YOLO segmentation model, generating the evaluation metrics that exposed YOLO's weaknesses on our data, and contributing to the model benchmarking that led the team to prioritize RF_DETR and SAM3.

Accomplishments: Marlyn contributed to prediction quality analysis and robustness evaluation for various models. She specifically trained a YOLO model we were considering before choosing RF_DETR and SAM3 for fish segmentation. After training the YOLO model, she created evaluation metrics, which exposed the disadvantages of YOLO and focused our efforts towards our other models.

- Trained and evaluated baseline YOLO segmentation models during early experimentation
- Generated evaluation metrics and visualizations exposing weaknesses in baseline YOLO performance
- Contributed to model benchmarking and comparisons that led to prioritizing RF_DETR and SAM3

Vibusha Vadivel: Primarily assigned to data preparation, preprocessing, and annotation-related workflows for the project. Her responsibilities include organizing and cleaning reef datasets, validating annotations, formatting datasets for training compatibility, and maintaining reproducible preprocessing pipelines. Vibusha also contributed to augmentation workflows designed to simulate underwater environmental conditions such as lighting variation, blur, and occlusion.

Expertise: Vibusha's has run controlled YOLOv8 experiments comparing random initialization, generic pretrained weights, and domain-specific pretrained weights on coral reef imagery. She has hands-on experience curating reef datasets from CoralNet, NOAA, and Scripps, including exporting data through CoralNet-Toolbox, structuring YOLO-formatted datasets with proper train/val/test splits, and handling both labeled and unlabeled data in a single pipeline. This reef-specific experience, combined with her general end-to-end Python ML pipeline skills, directly supports her ownership of the buoy dataset annotation, dataset pipeline documentation, and augmentation workflows for environmental validation.

Accomplishments: She organized and formatted reef datasets for training compatibility and improved our annotated dataset of images provided by our sponsor. She also assisted in creating a subset of buoy labelled images to aid in improving the accuracy of the segmentation models in recognizing non-fish objects. She contributed to the analysis and documentation of our system for usability purposes and to report findings for our project sponsors.

- Organized, cleaned, and formatted reef datasets for training and evaluation compatibility
- Documented workflows, dataset preparation procedures, and reproducibility findings for project sponsors and final reporting
- Contributed to the annotated reef fish and buoy datasets used during experimentation

MVP Completion

The final MVP showcases a working place-specific reef fish identification pipeline that combines segmentation and classification models. The core project goals and promised functionality were accomplished.

- **A working end-to-end Python package that can detect and segment fish in reef images completed**

Our original MVP proposed implementing a two-stage pipeline involving segmentation and classification, specifically a Segmenting Anything model, followed by a fine-tuned YOLO classifier. However, during development and testing, we identified a couple of models that excel at detecting and isolating objects in images for the pipeline segmentation component. We fulfilled the original MVP goal of fish segmentation and successfully took a reef image as input, segmented the fish, cropped the detected fish, and passed the cropped fish to the classifier.

- **Location-specific classifier completed**

Our original MVP suggested using a YOLO model as a classifier. However, during development and testing, we found that YOLO's performance wasn't strong enough for our purpose. Based on feedback and research, we switched to BioCLIP, which produced more accurate results for fish classification than YOLO. Per our sponsor's request, we focused primarily on three locations Moorea, Fiji, and Thailand. For each region, our system loads a top 25 species list into BioCLIP to classify fish only within that region. This maintains and fulfills the original goal of place-specific reef fish classification.

- **Evaluation completed using alternative datasets**

The original MVP planned to use the provided Moorea reef dataset as a concrete benchmark, but during evaluation and testing, we found significant issues with the annotations and labeling. Many images contained many visible fish, but only a subset of those fishes labeled, making the evaluation on this dataset somewhat unreliable. Since our project allowed alternative reef imagery, we found higher-quality labeled fish datasets on Kaggle and iNaturalist whose image quality is like that of the provided dataset. This provided us with a more realistic evaluation under similar conditions. It confirmed that the pipeline works on real, representative images and established a baseline accuracy.

- **Documentation and testing completed**

We completed documentation describing installation, dependencies, dataset setup, and how to run the full segmentation and classification pipeline. Unit tests cover individual components of the pipeline. Functional testing ensures that data and inference are working properly, as well as the full pipeline including segmentation and classification.

Milestone Completion

By the milestone checkpoint, the team had completed the core end-to-end reef fish workflow: fish localization, crop generation, region-constrained classification, model evaluation, and hard-negative dataset preparation. This revised section consolidates the original screenshots into concise tables and captioned figures so the results are easier to scan while preserving the technical evidence behind the milestone.

Completion Status

Milestone area	Status	Evidence of completion
End-to-end localization and crop generation	Completed	Produces boxes/masks, crops detected fish, and passes outputs to the classifier.
Place-specific species classification	Completed	BioCLIP uses region-specific candidate species lists for Moorea, Fiji, and Thailand.
Model comparison and evaluation	Completed	RF-DETR, SAM3, and BioCLIP evaluated on Coral Gardeners, DeepFish/Kaggle, and iNaturalist-style data.
Failure-case analysis	Completed / ongoing	Buoys and circular red objects identified as key false positives; hard-negative dataset completed, retraining of models started.
Reproducibility artifacts	Completed / ongoing	COCO-style annotations for dataset, image augmentation scripts, and model fine-tuning scripts added for reproducible improvement.

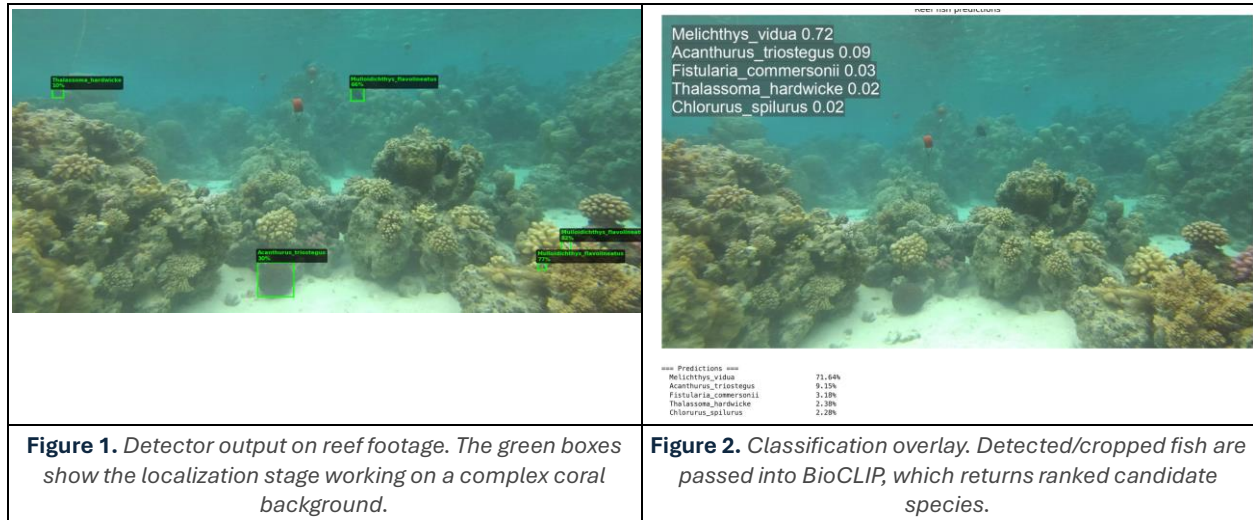
Key Evaluation Results

Model / dataset	Scale	Precision	Recall	F1 / accuracy	Main takeaway
YOLO 11	1,778 imgs; 250 boxes	6.89%	-	7.89%	Pivoted away after experimentation with YOLO 8, 11 and 12.
RF-DETR baseline	1,778 imgs; 5,969 det.	49.6%	41.5%	F1: 45.2%	High FP/FN baseline.
Detector v3	455 imgs; 1,472 boxes	60.39%	65.37%	F1: 62.78%	Stronger small test run.
SAM3 eval	274 imgs; 132 boxes	74.24%	77.17%	F1: 75.68%	Best segmentation signal.
BioCLIP CG	5,967 preds; 412 strict imgs	-	-	Norm. acc.: 49.64%	Real footage/domain shift.
BioCLIP iNat	107 samples	-	-	Acc.: 83.18%; w-F1: 84.46%	Cleaner upper-bound.

Visual Evidence and Model Outputs

Each image is paired with a short description explaining why it matters for the milestone.

Initial YOLO Model that Passes to BioCLIP



YOLO: The main takeaway after extensive experimentation is that YOLO does not generalize well enough across highly variable underwater environments. Since the Coral Gardeners use case involves placing a camera in different reef locations and detecting fish under changing conditions, the model was not robust enough for this task. This limitation is largely due to the lack of large, consistently labeled datasets that match the diversity of real Coral Gardeners' footage. As a result, YOLO performed reasonably in some controlled cases but struggled with domain shift, lighting changes, background variation, and inconsistent fish visibility. Because of these limitations, we pivoted away from YOLO and shifted our focus toward RF-DETR and SAM3, which showed stronger potential for the final pipeline.

Experimentation with Segmentation (RF_DETR vs SAM3)

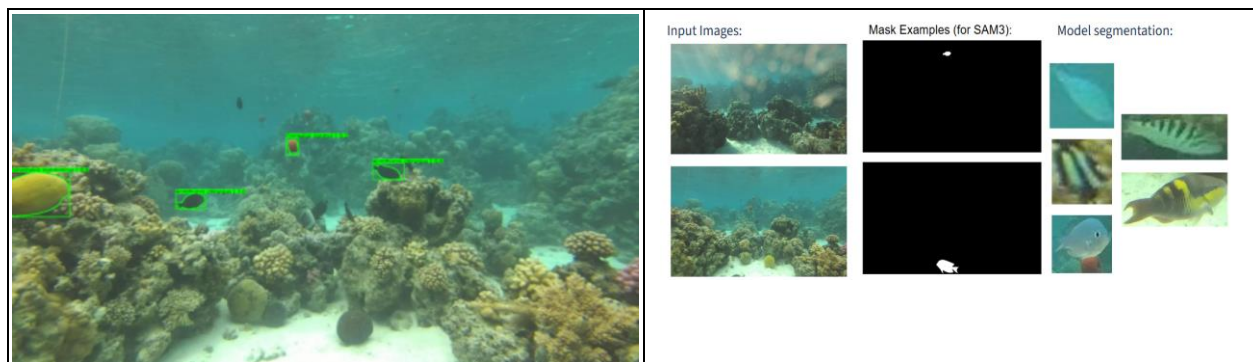
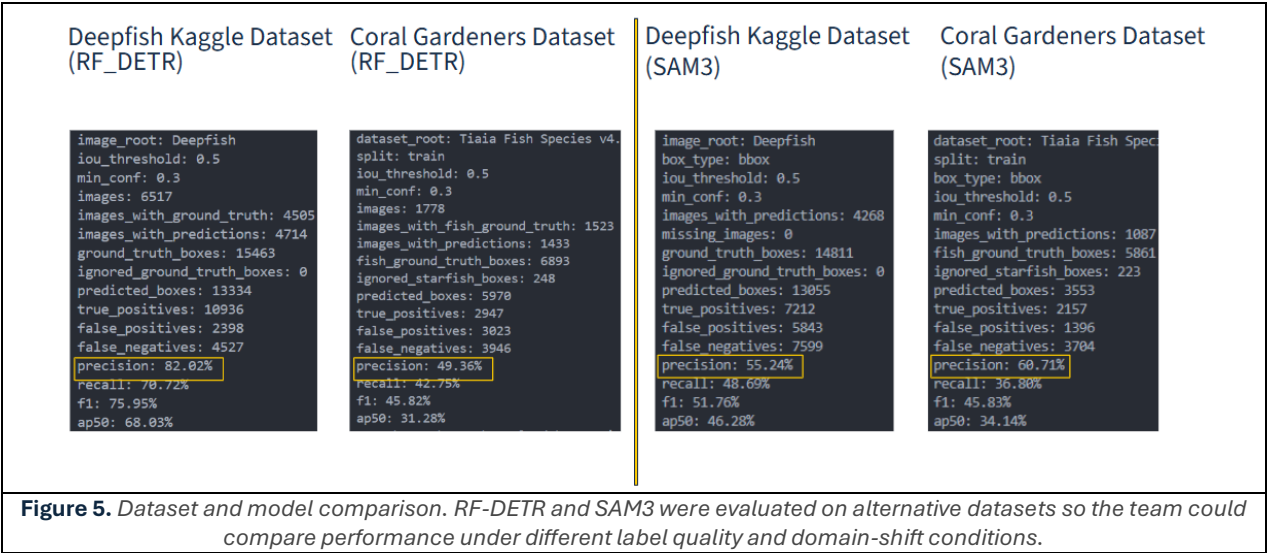


Figure 3. SAM3 visual run. This confirmed that the segmentation path could identify fish-like regions and generate candidate crops.

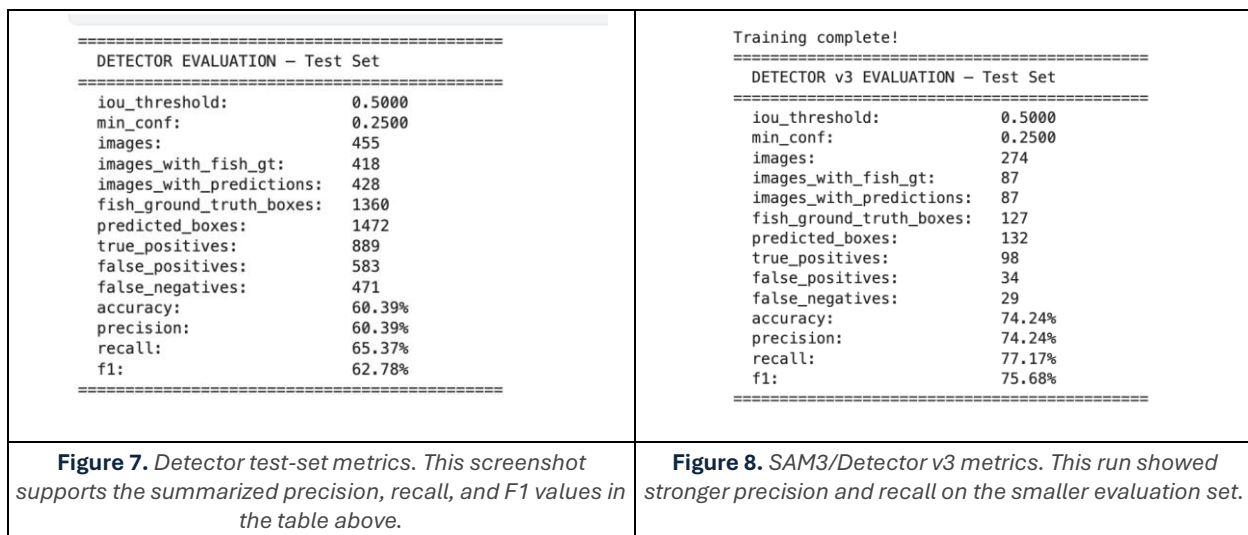
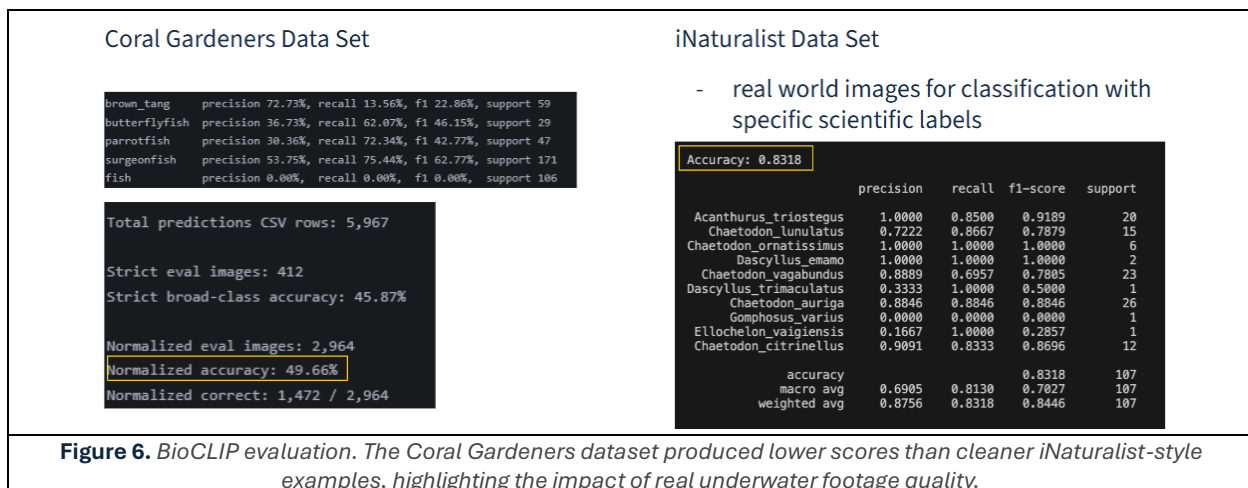
Figure 4. SAM3 input-to-output workflow. The figure shows source images, mask examples, and resulting fish crops for classification.



RF-DETR: The main issue observed with RF-DETR is that the model sometimes detects buoys as fish, which increases false positives and lowers overall accuracy. To address this, we created a negatively labeled dataset containing buoy examples. These samples will be treated as hard negatives during fine-tuning of the pre-trained model. By explicitly teaching the model to distinguish buoys from real fish, we expect to reduce false positives and potentially improve detection accuracy significantly.

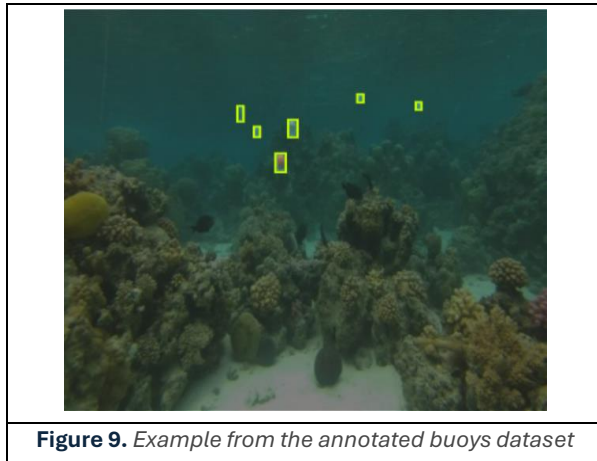
SAM3: It shows more consistent performance across datasets compared to the other segmentation approaches. One issue with the Coral Gardeners dataset is that not every visible fish is labeled in the ground truth annotations. Because of this, when SAM3 correctly detects an unlabeled fish, the evaluation may still count that prediction as a false positive, even though the detection is visually correct. This makes the reported precision and overall accuracy lower than the model’s actual performance. Moving forward, we plan to improve the ground truth annotations so that the model performance is reflected accurately. We also plan to fine-tune SAM3 through additional prompt engineering and experimentation with segmentation settings to increase precision and reduce false positives.

BioCLIP Integration in Pipeline



BioCLIP: BioCLIP classification on the Coral Gardeners dataset reached only 49.66% normalized accuracy across 2,964 evaluation images, compared to 83.18% on the cleaner iNaturalist dataset, confirming that real underwater footage quality is the bottleneck rather than the classifier itself. BioCLIP struggles most when crops are ambiguous or lack distinguishing features. Detector performance shares similar results, with the test-set evaluation showing 60.39% precision and 65.37% recall (62.78% F1) across 1,360 ground truth boxes, while the SAM3/Detector v3 run on a smaller 274-image subset reached 74.24% precision and 77.17% recall (75.68% F1). The gap between these two detectors runs suggests segmentation quality and evaluation set composition both affect downstream classification, reinforcing that improving crop quality (through buoy filtering, preprocessing the annotations, augmenting the images, and fine tuning the segmentation model) is best way to improve end-to-end pipeline performance.

Hard-Negative Dataset and Retraining Artifacts Milestones and Deliverables



To reduce false positives from buoys and other non-fish objects, the team added annotated buoy data and retraining support:

- [Buoy annotation data and COCO JSON for retraining](#) - supports explicit hard-negative examples for buoy false positives.
- [Dataset preparation and RF-DETR hard-negative fine-tuning scripts](#) - prepares the data layout and training workflow for reducing false detections.

Creating a dedicated buoy dataset was important because our current RF-DETR model had 3,023 false positives on the Coral Gardeners dataset, and buoys are a major driver of that error since they share visual features with fish (round shape, distinct color against blue water). Right now the model treats buoys as ambiguous background, sometimes calling them fish and sometimes not, which tanks precision to 49.36%. By labeling buoys as their own class, we plan to teach the mode much sharper decision boundary. This directly attacks the largest known failure mode without requiring more fish data.

Overall, the milestone was completed: the project now has a functional pipeline, baseline quantitative results, visual outputs, and a clear refinement path focused on false-positive reduction, dataset quality, and improved classification reliability.

Future Milestones

Week 8:

For week 8, our remaining milestones for our project are to focus on refining our end-to-end pipeline of reef fish identification, improving our evaluation metrics, and preparing our final deliverables. Since we have completed the segmentation and classification workflow, the

next step is to make our system more reliable, easier to use, improve accuracy, and reproducible.

We will address the main failure that we have identified during our testing, especially dealing with domain shift, low-resolution images, poor lighting, and camouflage. In order to improve performance and dealing with these image conditions, we will be working on doing image preprocessing, which include and are not limited to contrast enhancement, normalization, and other techniques that can make the fish be more visible before we segment and classify. We will do validation by going image by image and doing manual checks and verifications.

We have identified an issue within our segmentation model, that it sometimes detects buoys or other non-fish objects as fish. Since these objects do appear often in the Coral Gardeners footage, it can confuse our model. We are currently in the process of manually annotating images to capture buoys as a separate class. By doing this, it will allow us to train and evaluate the segmentation model more accurately and reduce the number of false positives we are obtaining.

We will also continue working on the region-specific species lists by collecting the top 25 fish species for Gulf of Thailand, Fiji, and Moorea. In addition, we will evaluate the Global Biodiversity Information Facility (GBIF) trained model compared to the iNaturalist-trained model to see which dataset gives stronger classification performance.

Week 9:

For week 9, we will delve deeper into comparing RF-DETR and SAM3 to determine which segmentation model works best for our fish detection in any type of image. While we are comparing the models, we will be getting segmentation metrics, such as false positives, false negatives, true positives, etc., getting error analysis, per-species prediction accuracy, and visual examples of the model's different segmentation on the same data set to see successful and failed detections. We want to be able to choose the strongest and most accurate segmentation model to fine tune it to get better results and accuracy for our final pipeline.

Additionally, we want to test different edge cases with different images, like blurry ones, having partial fish in the image, crowded environments, etc.

We will integrate the region-specific species lists into the BioCLIP classification pipeline. Instead of allowing BioCLIP to choose from a broad set of labels, we will limit predictions based on the selected region to its top 25 fish species, such as Gulf of Thailand, Fiji, or

Moorea. This will help the model make predictions based on fish species that are more likely to appear in the selected environment.

Week 10:

For our last week, we will then focus on usability and reproducibility. We will be improving the command-line interface, cleaning up the code to make it more readable and easier to understand, and organizing what is being outputted to make it easier for the user. We will also want to document how to run each of the models.

Lastly, we want to finalize the evaluation, demo, slides, and report for Coral Gardeners. Making it so that the final deliverables will include the strongest results possible for our segmentation and classification models, visualizations, and documentation where it explains the full end-to-end pipeline from having the images as our input and fish prediction as the output. Additionally, in this documentation, it will also talk about our failure cases that we had run into throughout the entire timeline.

Rest of Quarter Timeline

Our original project specification timeline for weeks 8 to 10 involved mainly finalizing the detection/classification pipeline. While we have achieved our MVP of building a working pipeline, there is room for improvement for fish species detection accuracy. As a result, we have adjusted our timeline from the original project specification to focus on iterating our system as suggested by our project sponsor. All team members are contributing to the final report and presentation steadily over the course of the remaining three weeks.

Week 8 (5/18 - 5/22)

Tasks

- **Alex:** Collect and research data on species frequency (top 25 LTER) for the three locations
- **Ewan:** Integrate and evaluate several preprocessing methods for experimentation, specifically for segmentation (RF_DETR, SAM3)
- **Marlyn:** Continue model refinement and evaluate preprocessing methods on classification (BioCLIP)
- **Vibusha:** Continue buoy dataset annotation, retrain segmentation models with hard-negative datasets
- **Kaleigh:** Train and evaluate model on GBIF data instead of iNaturalist and evaluate performance

Deliverables

- **Alex**: Documentation and writeup of species frequency for integration
- **Ewan**: Evaluation metrics for various preprocessing methods for segmentation (RF_DETR, SAM3)
- **Marlyn**: Evaluation metrics on various image preprocessing techniques on classification (BioCLIP)
- **Vibusha**: Dataset with buoy class and metrics of retrained segmentation models
- **Kaleigh**: Evaluation metrics for GBIF-trained model against iNaturalist-trained model

Week 9 (5/25 - 5/29)

Tasks

- **Alex**: Finalize end-to-end pipeline functionality and integrate region specific species identification
- **Ewan**: Finalize preprocessing workflow for segmentation and integrate chosen method into pipeline
- **Marlyn**: Conduct edge case testing and error analysis for final report
- **Vibusha**: Finish validation testing, complete documentation of dataset pipelines and Python package documentation
- **Kaleigh**: Finish segmentation evaluation and edge-case testing for GBIF model

Week 10 (6/1 - 6/5)

Tasks

- **Alex**: Ensure full system integration and document for reproducibility through a demo
- **Ewan**: Document preprocessing and dataset methodology
- **Marlyn**: Document comparative model analysis and evaluations/results
- **Vibusha**: Write dataset preparation and annotation workflow for reproducibility
- **Kaleigh**: Create segmentation visuals and improve usability through command-line interface implementation

Deliverables

- **Alex**: Final verified end-to-end integrated pipeline with demo
- **Ewan**: Preprocessing and dataset methodology flowchart and writeup
- **Marlyn**: Evaluation and results documentation and graphs
- **Vibusha**: Final dataset preparation and annotation workflow writeup
- **Kaleigh**: Final segmentation performance and usability writeup and results